

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

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Demystifying Amazon EKS networking

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Agenda

- Kubernetes networking tenets and model
- Amazon VPC CNI
- IP consumption optimization with Amazon EKS
- Network isolation strategies

Kubernetes networking tenets and models

Kubernetes networking tenets

- All containers can communicate with all other containers without NAT
- All nodes can communicate with all containers (and vice versa) without NAT
- The IP address that a container sees itself as is the same IP address that others see it as

Container Network Interface (CNI)

- Network interface specification for Kubernetes
- Assign a network namespace and a network interface to the Pod at startup
- Assign an IP address to a Pod
- Clean up the network namespace when the Pod terminates, frees up the IP address

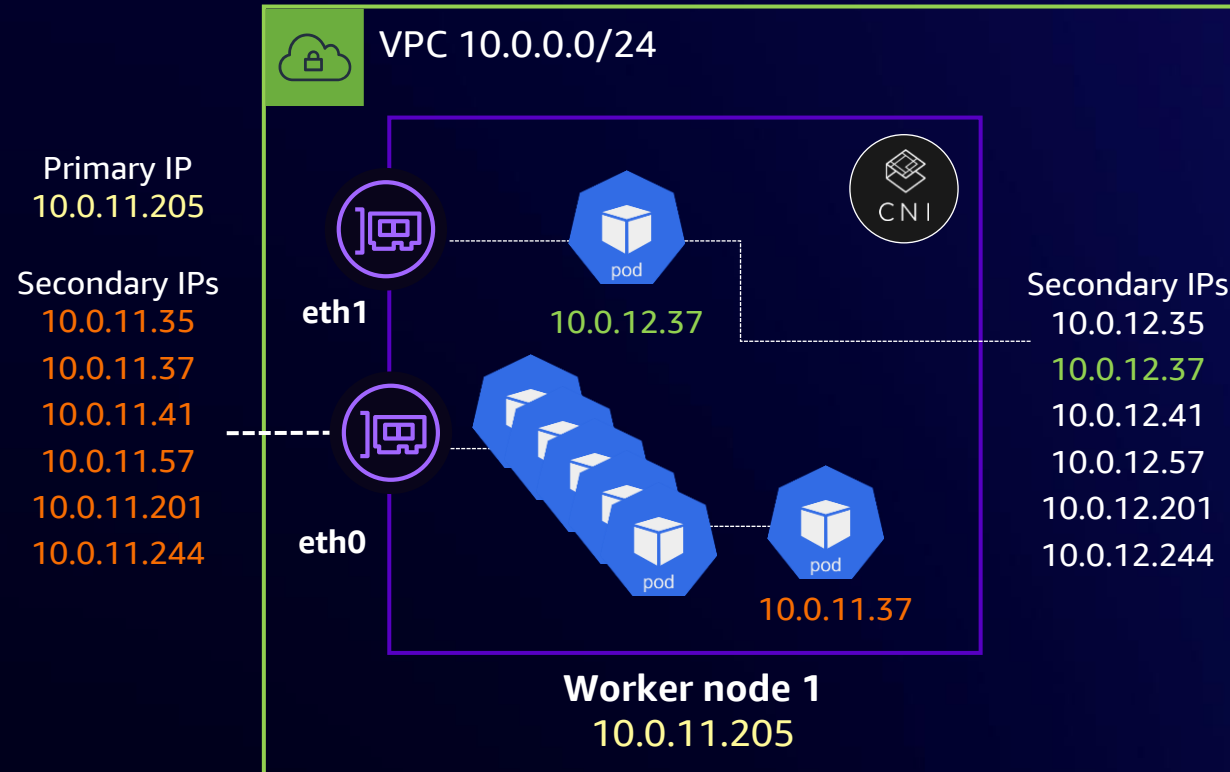


Amazon VPC CNI



EKS Pod networking

WITH AWS VPC CNI PLUGIN



Benefits

- **Reduced complexity:** Integrates with Amazon VPC networking
- **Improved performance:** No Network translation required
- **Security:** Integrates with AWS security features
- **Flexibility:** Highly customizable CNI

How it works

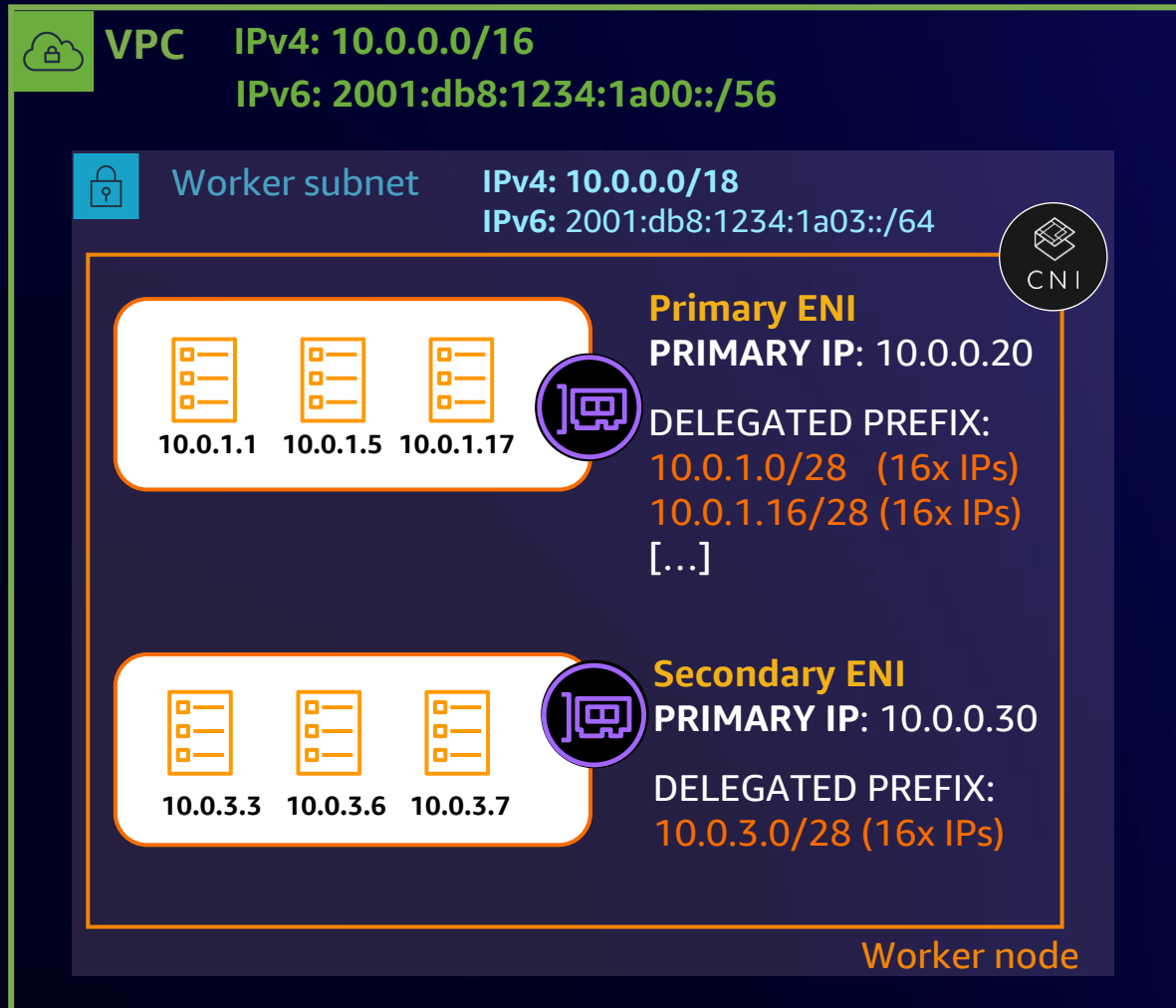
- Keeps a **warm pool** of secondary IP addresses to assign to pods
- **Attaches additional ENIs** when IP addresses on active ENIs are exhausted

IP consumption optimization with Amazon EKS



Optimize node utilization

EC2 PREFIX DELEGATION



Benefits

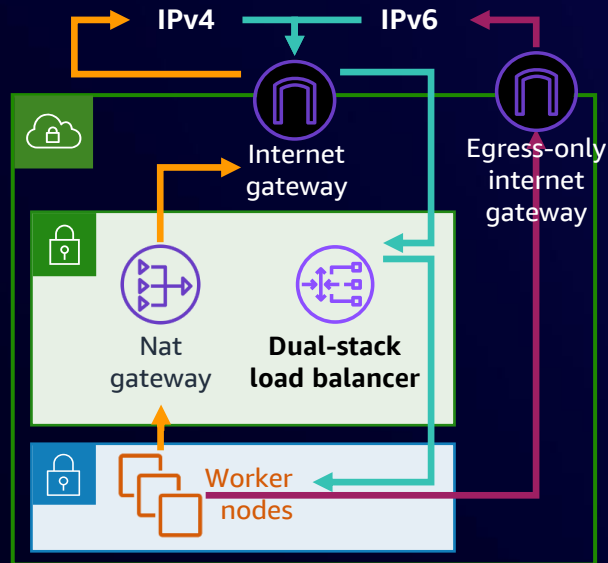
- **Improved efficiency:** Better instance utilization, increased Pod density, and fewer API calls to the EC2 control plane
- **Reduce costs:** By optimizing instance utilization
- **Flexibility:** Works with both IPv6 and Custom Networking

How it works

- Each ENI is assigned a **/28 prefix**
- **Pods are assigned IP addresses** from these prefixes

Increase address space

IPv6

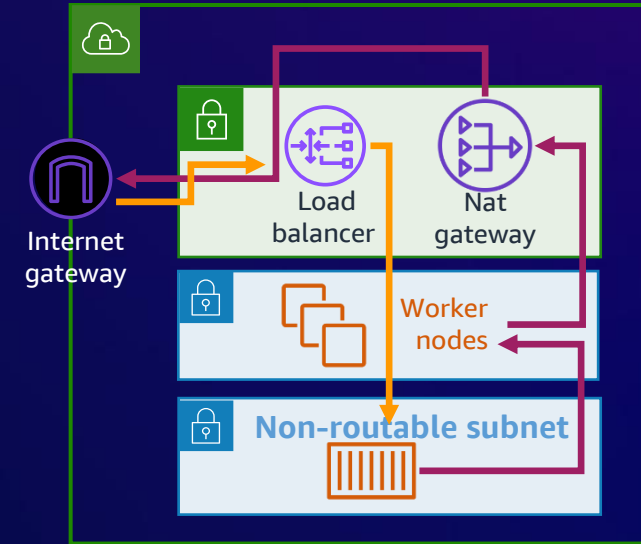


Connects to legacy applications: IPv4 endpoints can connect to IPv6 services and vice versa

Avoids IP exhaustion:

- 2^{64} IPv6 addresses (VPC CIDR block size of /64)
- Each worker node will be assigned a /80 IPv6 prefix to be used by the pods on that node

CUSTOM NETWORKING



Conserves routable IPs: Schedule Pods in dedicated CG-NAT subnets (100.64.0.0/10 or 198.19.0.0/16)

Extend pod subnets: Deploy an ENIConfig custom resource for pod subnets

Network isolation strategies

Kubernetes network policy

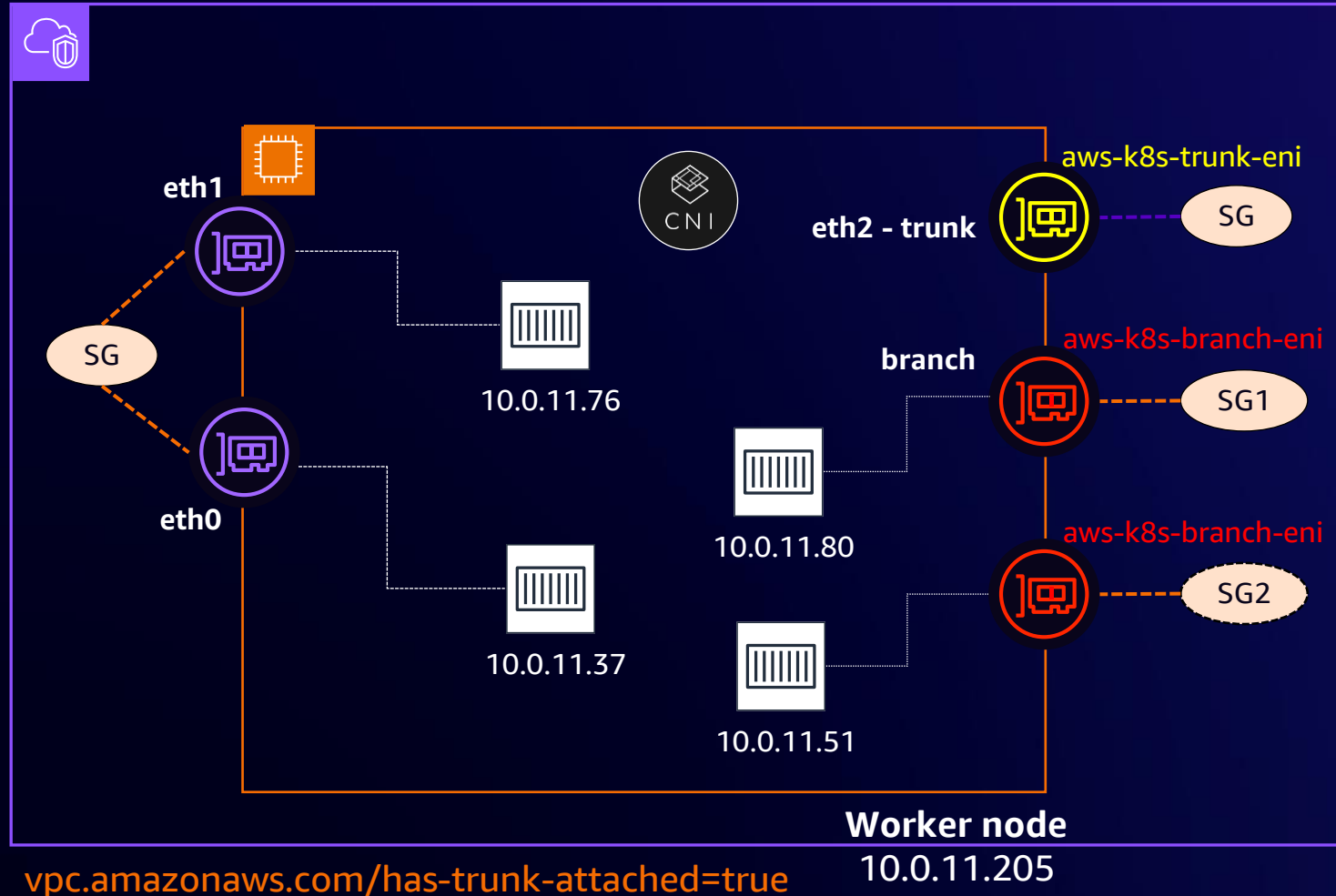


Security. Secure intra-cluster network traffic using Kubernetes network policies

Simplicity. Kubernetes network policies are application centric and help achieve pod isolation

Granularity. Kubernetes network policies identify three sets of entities – Pods, namespaces, IP blocks (CIDR ranges)

Security groups for Pods



Granular control. Security rules for individual Pods

Simplified management. Easily apply security rules to all of the Pods with common access needs at once

Improved compliance. Implement security standards and regulations

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